

CHARUSAT UNIVERSITY OF SCIENCE AND TECHNOLOGY
M. Sc. BIOCHEMISTRY (SEMESTER II)
EXTERNAL EXAMINATION
BC 708- HUMAN PHYSIOLOGY

TOTAL MARKS:20

DATE: 1.05.2010
TIME: 10.00 AM TO 1.00 PM

MARKS OBTAINED IN PART A:

SIGNATURE OF THE SUPERVISOR

TIME DURATION: 30MINS (10.00-10.30)

(PART A)

1. Lymphocytes

- a) originate from the bone marrow after birth
- b) are unaffected by hormones
- c) interact with eosinophils to produce platelets
- d) are part of body's defense against cancer

2. Which of the following is an important buffer in interstitial fluid?

- a) hemoglobin
- b) carbonic acid
- c) H_2PO_4
- d) compounds containing histidine

3. A hematocrit of 41% means that in the sample of blood analysed

- a) 41% of hemoglobin is in the plasma
- b) 41% of total blood volume is made up of blood plasma
- c) 41% of the total blood volume is made up of red blood cells
- d) 41% of formed elements in blood are red blood cells, white blood cells and platelets

4. In normal human blood

- a) eosinophil is the most common type of white blood cell
- b) there are more lymphocytes than neutrophils
- c) iron is mostly in hemoglobin
- d) there are more white cells than red cells

5. Which of the following are incorrectly paired?

- a) B cells: insulin
- b) D cells: somatostatin
- c) A cells: glucagons
- d) F cells: gastrin

6. Glucagon is not normally found in

- a) Brain
- b) pancreas
- c) plasma
- d) adrenal glands

7. Insulin increases the entry of glucose into

- a) skeletal muscle
- b) all tissues
- c) renal tubular cells
- d) most neurons in cereb
cortex

8. Which of the following hormones exerts the least effect on growth?

- a) Vasopressin
- b) testosterone
- c) T_4
- d) insulin

9. In human males, testosterone is produced mainly by the

- a) Leydig cells
- b) Sertoli cells
- c) seminiferous tubules
- d) vas deferens

10. Which of the following acts directly on the nucleus to produce physiologic effects?

- a) $\text{IL-1}\beta$
- b) Growth
- c) hormone dopamine
- d) estradiol

11. One of the following hormone increases muscle mass
a) Epinephrine b) testosterone c) thyroid hormones d) cortisol
12. What is the name of the muscular contractions which push food through the system?-----
13. The components of bile include: bile salts, bile pigments, cholesterol and various electrolytes. Which of the components in bile have digestive functions?
a) Bile salts b) Bile pigments c) Cholesterol d) all of the above
14. What enzyme produced by the kidneys helps to regulate blood pressure?
a) Aldosterone b) creatine c) Renin d) Erythropoietin
15. Which of the following is NOT a function of the kidneys?
a) maintain water and electrolyte balance b) maintain acid-base balance
c) produce rennin d) produce urea
16. Given the following conditions: Capsular hydrostatic pressure = 15 mm Hg
Glomerular plasma osmotic pressure = 25 mm, Glomerular hydrostatic pressure = 75 mm Hg. What is the NET FILTRATION RATE?
a) 35 mm Hg b) 85 mm Hg c) 65 mm Hg d) 115 mm Hg
17. During reabsorption, Na^+ crosses the apical walls of the proximal tubule cells into the interstitial fluid principally by means of:
a) Phagocytosis b) Receptor-mediated endocytosis c) Simple diffusion d) Active transport
18. The system that controls smooth muscle, cardiac muscle, and gland activity is the:
a) Somatic nervous system b) Autonomic nervous system c) Skeletal division
d) Sensory nervous system
19. Axons within nerves may have which of the following associated with them?
a) Schwann cells Nodes of Ranvier
b) c) Oligodendrocytes
c) A and b
d) All of the above
20. Action potentials are conducted more rapidly in:
a) Small diameter axons than large diameter axons
b) Large diameter axons than small diameter axons
c) Unmyelinated axons than myelinated axons
d) Axons that lack a wrapping of Schwann cells

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TOTAL MARKS: 50
DURATION: 2HR 30MINS (10.30-1.00)

(PART B)

Section I

- | | |
|---|---|
| Q1.(A) Write a comprehensive account on plasma proteins | 4 |
| Q1.(B) Write a general account on any one:
thalassemia
metabolic acidosis | 3 |
| Q1.(C) Briefly describe Intracellular buffer systems | 3 |
| Q2.(A) Write a short note on synthesis of thyroid hormones. | 4 |
| Q2.(B) Give a brief note on any two:
Hormone receptors
Renin-Angiotensin-Aldosterone pathway
Central <i>Diabetes insipidus</i> | 6 |

Section II

- | | |
|--|---|
| Q3.(A) Explain the control of gastric acid regulation | 3 |
| Q3.(B) Write down a brief account on functions of liver and its disorders | 4 |
| Q3.(C) Write the composition of pancreatic juice. | 3 |
| OR | |
| Explain the digestion and absorption of protein. | 3 |
| Q4.(A) Enlist the functions of Urinary system | 4 |
| Q4.(B) Write a note on Nephron | 3 |
| Q4.(C) Briefly explain the mechanism of urine formation. | 3 |
| OR | |
| Explain the action of Diuretics | 3 |
| Q5.(A) Explain structure & types of Neurons. | 3 |
| Q5.(B) Describe the physiology of muscle contraction | 3 |
| OR | |
| Write a note on Types of Muscles | 3 |
| Q5.(C) Define Neurotransmitter. Add a note on the role of Acetylcholine in human body. | 3 |